

The Solar System II

AS WELL AS THE PLANETS, THE GRAVITY OF THE SUN ATTRACTS A HUGE NUMBER OF SMALLER OBJECTS.

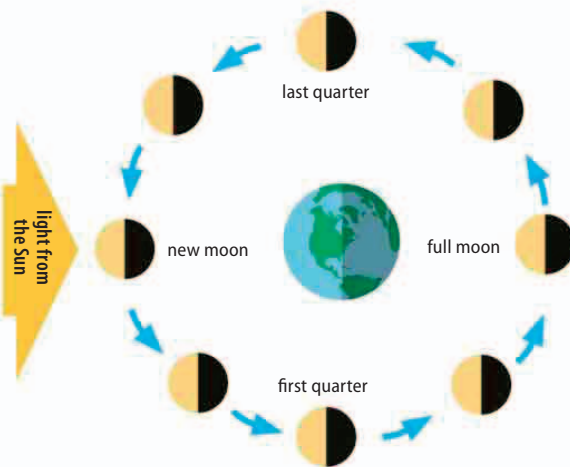
Moons orbit most of the planets, including Earth, while smaller bodies, such as comets, follow independent paths.

The Moon

A moon is a body that orbits a planet and there are more than 160 in our Solar System. Earth has just one moon, a cratered ball of rock, which spins as it orbits. The Moon formed when a large asteroid collided with Earth during its formation and some of the debris went into orbit around the Earth, becoming the Moon.

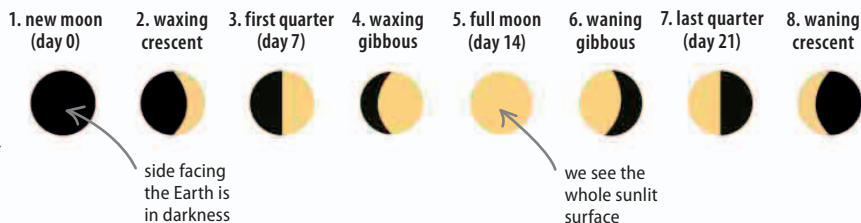
Orbit of the Moon ▷

When we look at the Moon, only the sunlit part is visible. The Moon takes the same amount of time to orbit our planet as to spin once on its axis, so we only ever see one side from Earth. The side we don't see is called the far side of the Moon.



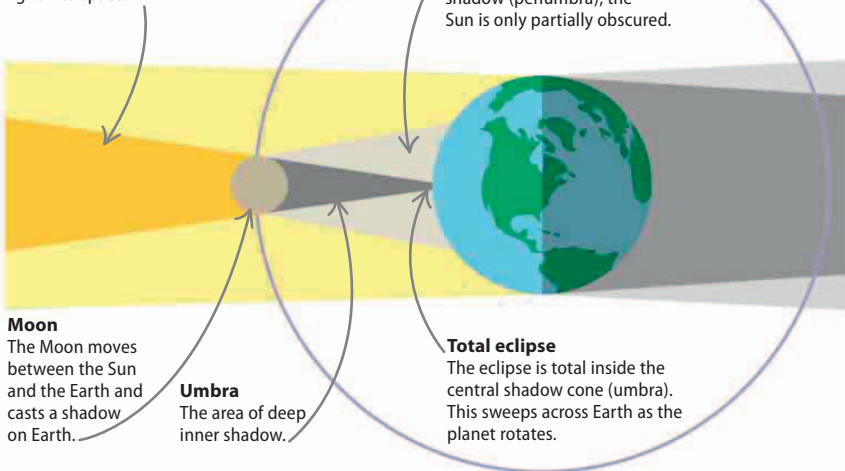
The lunar cycle ▷

Every month the face of the Moon appears to change from a dark shadow, called a new moon, to become a thin, shining crescent, to a full moon, and then back again. This is because, as the Moon orbits Earth, we see more or less of the half of the Moon that is lit by the Sun.



Sunlight

The Sun radiates light into space.



Eclipses

Sometimes, as the Moon orbits Earth, it passes directly in front of the Sun and blocks out the light. This is called a solar eclipse. Sometimes the Moon's path takes it into Earth's shadow, causing a lunar eclipse. When this happens the Moon dims and appears red because the light is bent as it passes through Earth's atmosphere.

◁ Solar eclipse

There are only about three total solar eclipses each year, and each one is only seen from the narrow band of deep shadow, called the umbra, on the Earth's surface. More frequent is a partial solar eclipse, when the Moon's shadow covers just part of the Sun.

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